**Thadomal Shahani Engineering College, Bandra**
**Department of Artificial Intelligence & Data Science**

| Name: Vansh Matlani | | | | Subject: SEPM Laboratory |
|---|---|---|---|---|
| TE- 1 | Batch: T13 | Sem: 6 | Roll No: 2301069 | Date: 19-01-2026 |

**Experiment No: 02**

**Aim:** To understand Version Control System / Source Code Management, install git and create a GitHub account

**Theory:**

- Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. Git is easy to learn and has a tiny footprint with lightning fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.

- Some of the basic operations in Git are:

    1. Initialize
    2. Add
    3. Commit
    4. Pull
    5. Push

- Some advanced Git operations are:
    1. Branching
    2. Merging
    3. Rebasing

The following diagram depict the all supported operations in GIT

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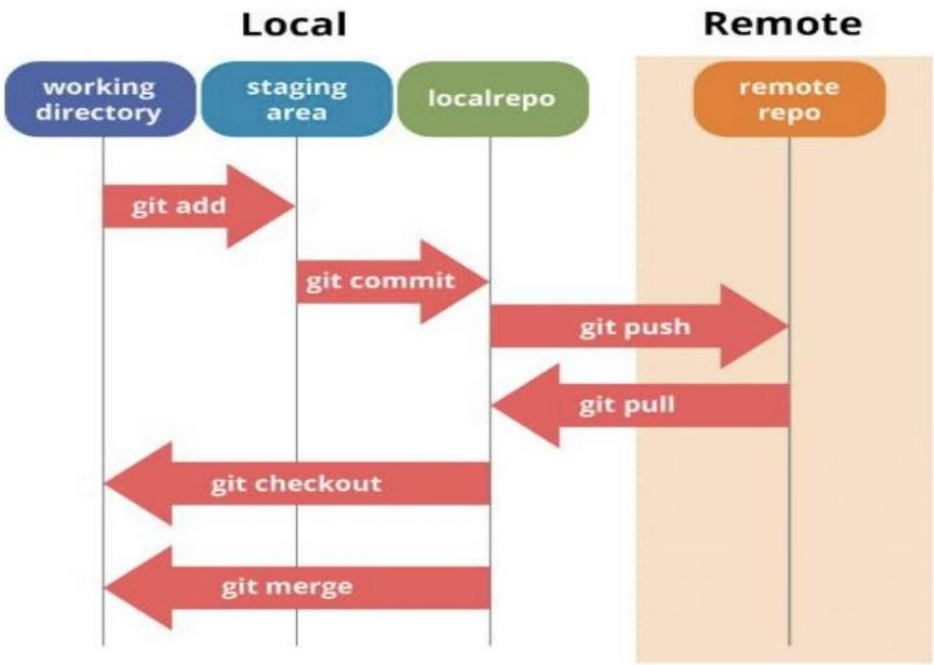


- **Installation of GIT**

1) In windows, download GIT from https://git-scm.com/ and perform the straightforward installation.

2) In Ubuntu, install GIT using $sudo apt install git,

Confirm the version after installation using command $git –version

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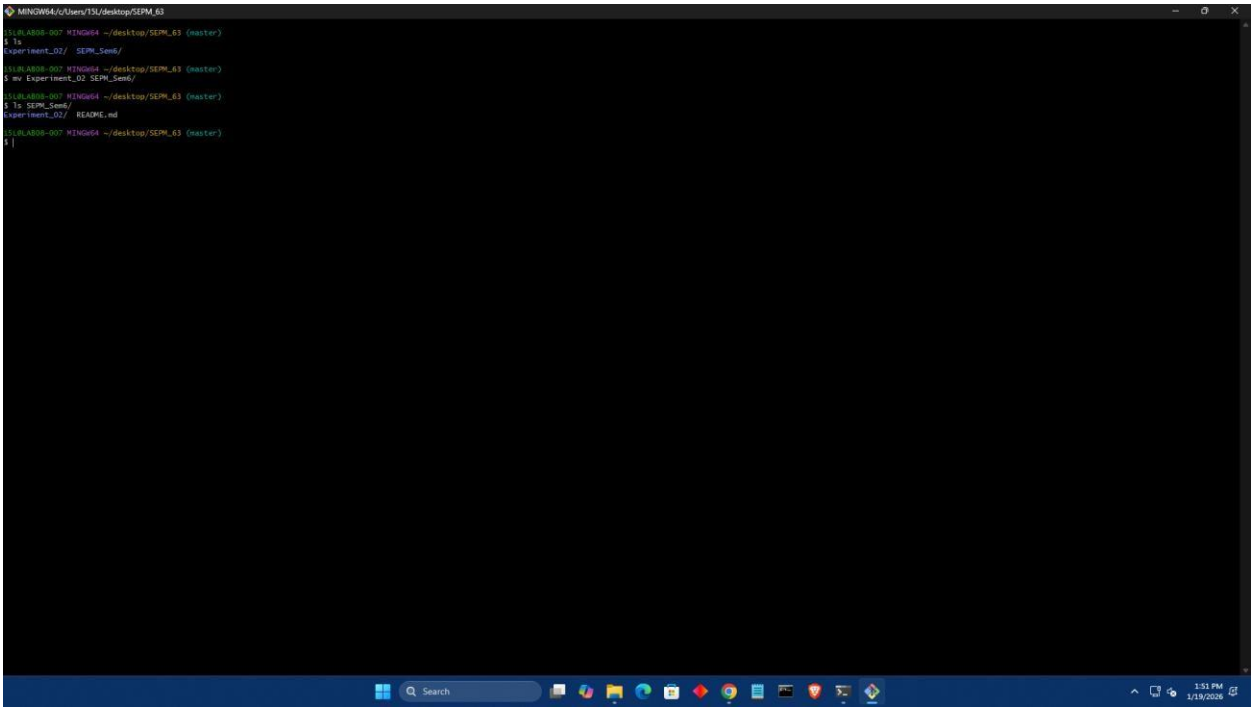
**Output:**



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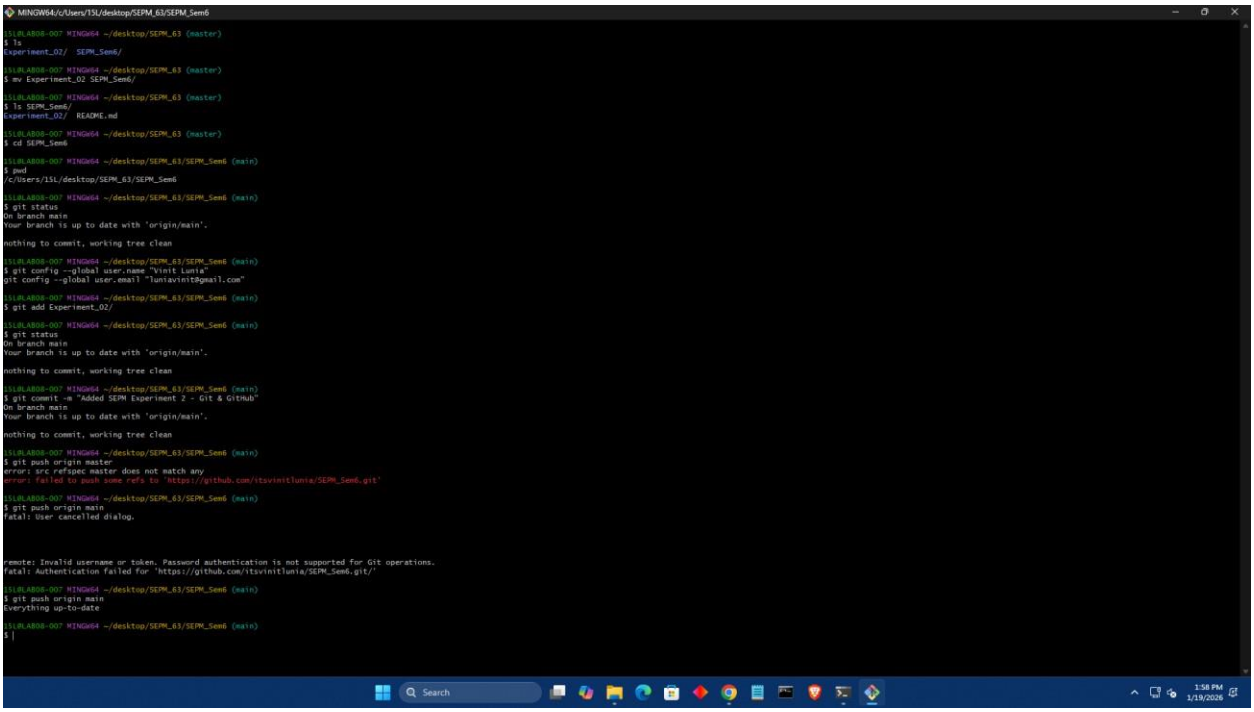


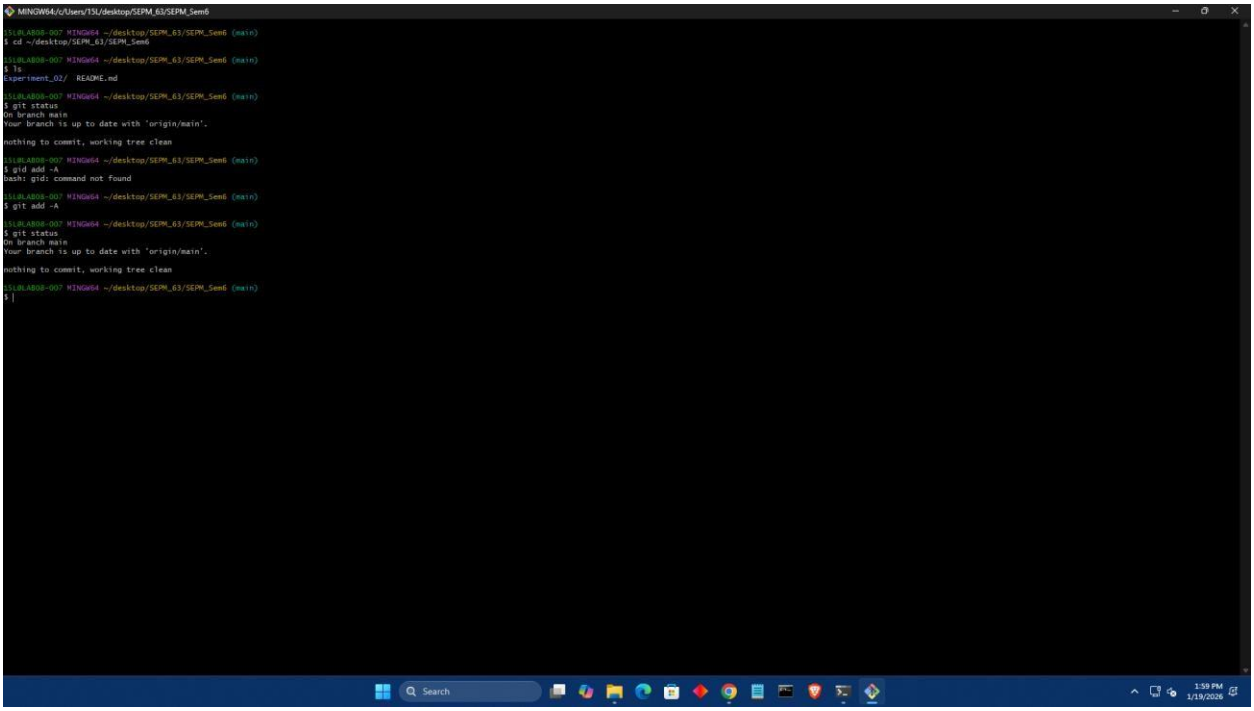

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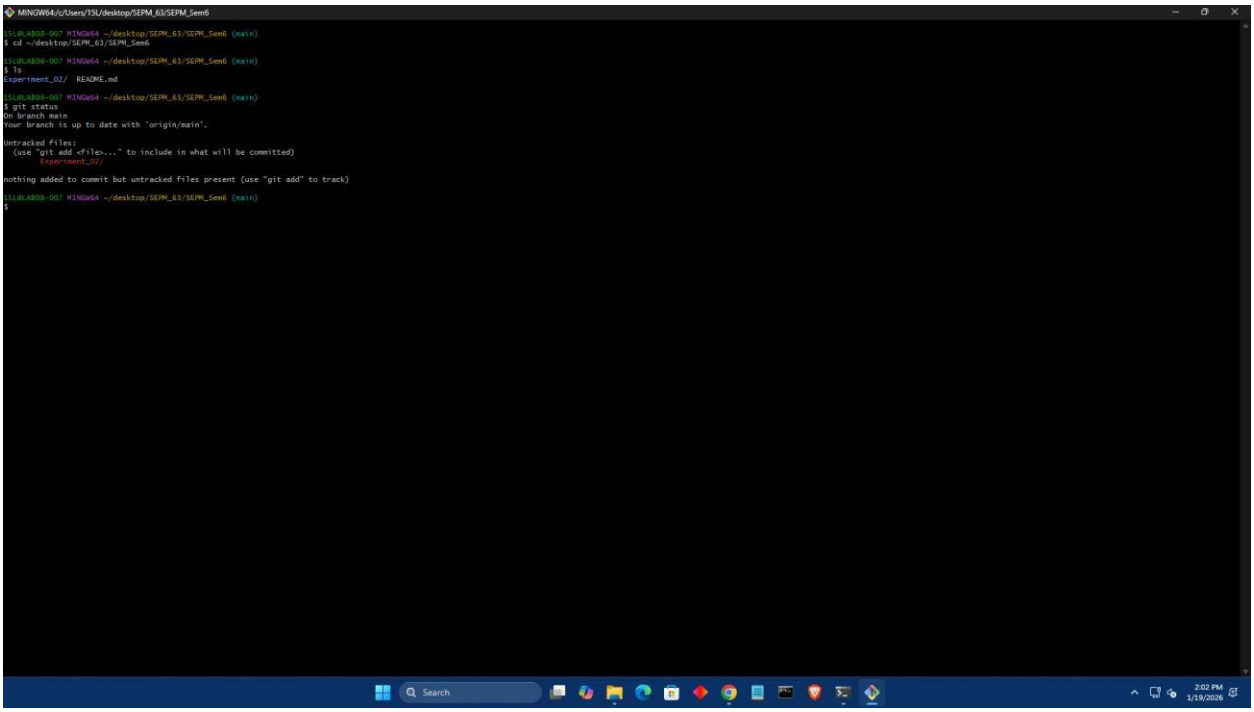


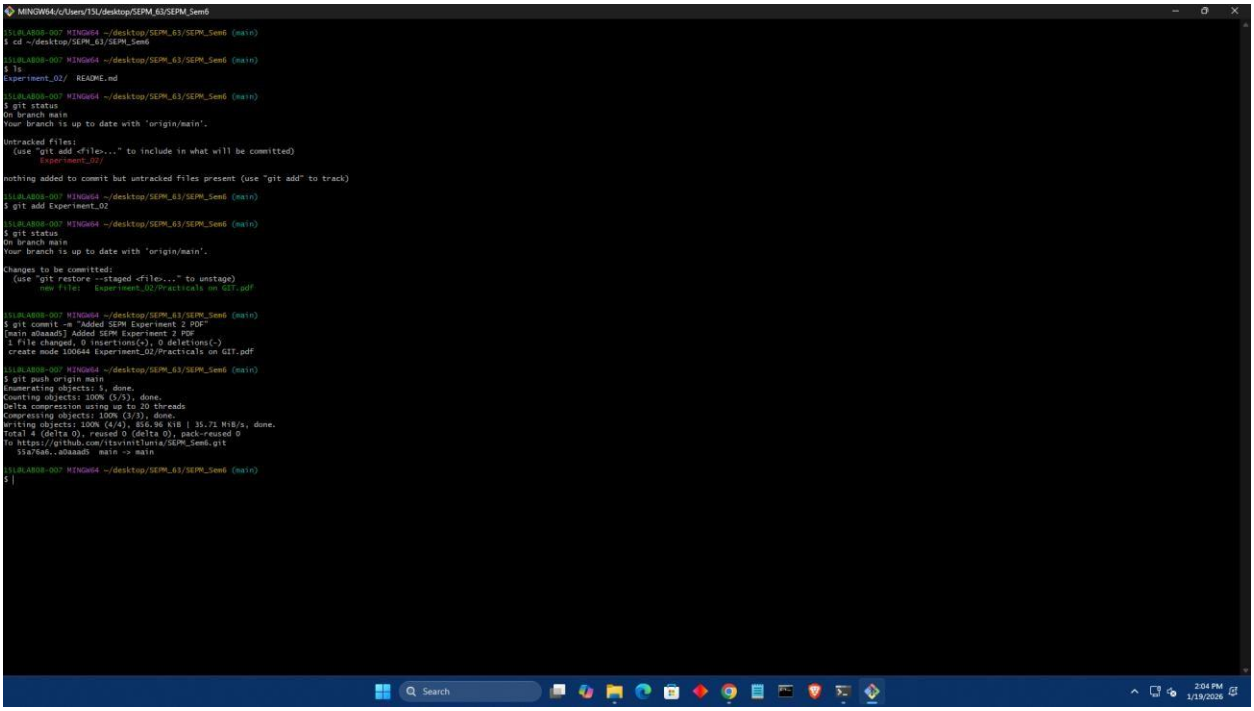

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**Conclusion:** Successfully performed Version Control System / Source Code Management, install git and create a GitHub account